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熊峻嶙

xiongjunlin1992@gmail.com
www.github.com/Xiong-Junlin
微信：xx781420201　　电话：18080663775
接受国内外中长期外派

个人简介

- 拥有 8 年解决方案与产品研发经验，深耕 5G 与智慧交通，聚焦 AI 等新兴技术在跨行业的研究与应用，善于将业务需求转化为落地的智能化解决方案。
- PMP 认证专家，具备丰富的项目管理与交付经验，熟悉从需求分析到实施的全流程，确保复杂项目成功落地。
- 精通 Python 及主流机器学习框架（PyTorch、Scikit-learn），深入理解算法与行业数据特征，可为 AI 产品设计与研发提供可行性评估与技术指导。
- 拥有欧洲研究与工作经历，**英语可作为工作语言**，具备在跨国团队协作的能力。

职业发展概览

2023.09–2025.08	阿尔托大学	双硕士
2025.03–2025.08	西班牙光子科学研究所	机器学习工程师
2020.07–2023.08	中国移动（上海）产业研究院	新技术解决方案经理
2018.06–2020.06	春秋航空	产品经理
2015.07–2018.06	深圳航空	产品经理
2011.09–2015.07	西南交通大学	本科

教育背景

- 2023.09-2025.08　阿尔托大学(Top 200)　芬兰
  布达佩斯技术与经济大学　匈牙利
  双硕士：自动化系统 - 机器人与人工智能
- 2011.09-2015.07　西南交通大学　中国
  本科：　测控技术与仪器

职业认证

产品研发	PDMA - 新产品开发专业认证（NPDP）
项目管理	PMI - 项目管理专家专业认证 (PMP)

工作经历

2025.03—2025.08 西班牙光子科学研究所

机器学习工程师

主要职责

- 作为机器学习研究员进行硕士论文实习，致力于开发高效的深度学习算法，提升计算效率与预测精度，应用于实时分子传感与智能光谱学。

项目经验

- **多模深度学习融合的智能光谱分析**：设计并实现了完整的傅里叶变换红外光谱（FTIR）分析流程，包括合成数据集生成、预处理与监督学习建模。利用多层感知机（MLP）与卷积神经网络（CNN）等多种模型进行基于Fisher信息量的多模融合，提出一个实时切准确的智能光谱分析系统。

2020.07—2023.08 中国移动（上海）产业研究院

新技术解决方案经理

主要职责

- 研究5G网络技术在民航飞行过程中的应用，重点聚焦驾驶舱飞行品质监控、飞机健康管理、客舱危险行为预警及智慧客舱系统的产品研发。
- 作为项目负责人，牵头对接行业合作伙伴，管理大型智慧交通项目，提供全方位的5G技术支持与行业解决方案。
- 定期输出基于5G网络的智慧交通创新应用场景，撰写并提交相关技术专利。

项目经验

- **基于移动智能设备的颠簸预测**：开发了一套基于便携式机载设备的实时晴空颠簸（CAT）预警系统，利用加速度传感器采集客舱实时数据，并通过地面端机器学习模型进行预测分析。在不依赖额外硬件的前提下，显著提升飞行安全性与航班调度效率。
- **东方航空机上5G平台项目**：主导中国东方航空5G地空互联方案的定制开发与实施，搭建网络管理与业务数据分析平台。基于机上实时数据，实现了旅客行为识别、动态服务推荐与危险行为预警。带领民航大学研发团队完成平台从设计、开发到交付，显著提升了航旅服务智能化水平。
- **中国商飞C919试航认证项目**：协调中兴通讯、中国商飞及多家合作单位，统筹认证流程、网络设备部署与项目进度管理。作为行业技术专家，提供**“5G+航空”融合解决方案**，确保C919试飞任务的高效推进与合规交付，助力国产大飞机迈向商用化阶段。

2018.06—2020.06 春秋航空 产品经理

主要职责

- 负责业务流程优化，识别关键痛点，推动IT系统标准化建设，提升整体运营效率。
- 定期进行行业趋势分析与竞争对标研究，协助制定公司中长期发展战略，推动数据驱动决策，实现IT新技术在民航业的解决方案。

项目经验

- **EFB电子飞行包项目**：作为项目负责人，成功推动EFB系统落地，实现无纸化放行，成为中国民航业首例。该系统显著提升飞行前安全检查与航班变更的应急响应效率，每年为公司节省超过**150万元**。
- **飞行运行质量分析项目**：主导设计并建设飞行运行质量分析平台，从机场、航线、航班等多个维度开展智能分析，首次实现公司、机场与空管三方质量数据的深度融合。该系统使飞行质量**效率提升超过20%**。
- **中国商飞航班运行项目**：为商飞试飞中心量身定制航班运行管理平台，涵盖智能排班、飞行放行、航班监控及质量评估全流程。系统通过数据智能化驱动飞行品质分析，显著提升试飞任务的执行效率与质量管控水平，获得客户高度认可。
- **春秋航空 WMS 项目**：构建覆盖全流程的仓储管理系统（WMS），涵盖从仓库到机上消耗品配送、机上销售与回收全过程。系统实现了销售模式驱动的智能交接、实时追踪及自助库存查询，有效优化仓储、出入库及对账流程，显著提升了供应效率与准确性。

2015.07—2018.06 **深圳航空** **产品经理**

主要职责

- 作为项目经理监控项目进度与质量，确保需求开发的时间，成本，质量三要素。管理研发团队，开展业务调研工作并提供业务与技术的持续支持。
- 独立完成产品从0到1的开发，并推进产品推广工作。

项目经验

- **CDM机场运行平台**：负责机场运行数据的对接和集成管理，为后期航班协调决策提供数据支持。
- **FOC航班放行平台**：为客户提供监控放行流程、航班状态与异常报警等功能开发和优化，为客户提供解决方案。

研究项目

多模态视觉感知与语义决策的智能驾驶系统 **2025.06 - 至今**

为应对智能驾驶场景下对目标识别与语义决策的高实时性与准确性需求，构建了基于YOLOv8 + SORT + 本地大语言模型（LLM）的多模态视频分析系统。系统支持对行车视频中多类目标（如行人、车辆、交通标志）进行精准检测与轨迹追踪，并融合语义理解模块，实时生成驾驶辅助提示信息，有效提升智能驾驶系统的环境感知与决策能力。

基于扩展卡尔曼滤波器的GPS与IMU传感器融合定位 **2024.09 - 2024.11**

针对单一传感器定位系统在动态环境下精度不足问题，设计了融合GPS与IMU的扩展卡尔曼滤波器（EKF）定位模型。基于Python语言和KITTI数据集实现车辆轨迹模拟与路径估计，显著提升系统在非理想条件下的稳定性与精度，为高精度导航与自动驾驶提供可行性验证。

基于RRT算法的移动机器人路径规划 **2024.05 - 2024.08**

为解决复杂环境中移动机器人路径规划问题，采用RRT算法构建无碰撞路径规划系统，并在Webots仿真平台完成仿真实现。项目聚焦于障碍物规避与路径优化，适用于多种非结构化或动态变化场景，有助于提升机器人在实际部署中的通行效率与自主决策能力。

XIONG JUNLIN

xiongjunlin1992@gmail.com
www.github.com/Xiong-Junlin
WeChat：xx781420201 Phone：+86 18080663775

Open to international assignments

Summary

- ***8 years of experience in solution design and product development,*** *with a strong focus on 5G and smart transportation. Specialized in applying emerging technologies such as AI across industries, with expertise in translating business requirements into practical intelligent solutions.*
- ***PMP-certified project management professional,*** *experienced in end-to-end delivery from requirements analysis to implementation, ensuring the successful execution of complex projects.*
- ***Proficient in Python and leading machine learning frameworks (PyTorch, Scikit-learn),*** *with deep understanding of algorithms and industry data characteristics. Skilled in conducting feasibility assessments and providing technical guidance for AI product design and development.*
- ***International academic and professional background in Europe,*** *fluent in English as a working language, with proven experience collaborating in cross-cultural and multinational teams.*

Career Summary

2023.09–2025.08	Aalto University	M.Sc.
2025.03–2025.08	Institute of Photonic Sciences, Spain	Machine Learning Engineer
2020.07–2023.08	China Mobile Industrial Research Institute	Solution Manager
2018.06–2020.06	Spring Airlines	Product Manager
2015.07–2018.06	Shenzhen Airlines	Product Manager
2011.09–2015.07	Southwest Jiaotong University	BEng.

Education

- 2023.09-2025.08 Aalto University Finland
  BME University Hungary
  M.Sc. Autonomous Systems - Robotics and Artificial Intelligence
- 2011.09-2015.07 Southwest Jiaotong University China
  BEng. Measurement and Control Technology and Instruments

Certification

Software Product	**PDMA** - New Product Development Professional Certification
Project Management	PMI Project Management Professional (**PMI-PMP**)®

Professional Experience

2025.03—2025.08 **The Institute of Photonic Sciences (ICFO)**
Machine Learning Engineer

Responsibilities

- Conducted master's thesis research internship at ICFO, one of the world's leading photonics research centers. Focused on developing efficient deep learning algorithms to enhance computational performance and predictive accuracy for real-time molecular sensing and intelligent spectroscopy applications.z

Project Experience

- **Intelligent Spectral Analysis Based on Deep Neural Networks:** Designed and implemented a complete pipeline for FTIR spectral data, including synthetic dataset generation, preprocessing, and supervised learning. Developed multivariate regression models using Scikit-learn and PyTorch, covering both traditional machine learning models and deep neural networks (MLP, CNN). Conducted extensive experimentation and comparison of model performance, regularization, and generalization through learning curves and cross-validation.

2020.07—2023.08 **China mobile (Shanghai) Industrial Research Institute**
Solution Manager

Responsibilities

- Conduct research on the application of 5G network technologies in civil aviation, with a focus on cockpit flight quality monitoring, aircraft health management, in-cabin abnormal behavior detection, and smart cabin system development.
- Serve as project lead, liaising with industry partners and managing large-scale smart transportation projects, while providing comprehensive 5G technical support and customized industry solutions.
- Regularly deliver innovative smart transportation application scenarios leveraging 5G, and author relevant technical patents.

Project Experience

- **5G In-Flight Platform Project – China Eastern Airlines:** Led the customized development and implementation of a 5G ATG solution for China Eastern Airlines. Developed a network management and business data analytics platform to support real-time data processing onboard. Achieved passenger behavior recognition, dynamic service recommendation, and abnormal behavior alerts.
- **C919 Flight Test Certification Project:** Coordinated efforts among ZTE, COMAC, and multiple partner organizations to manage the certification process, network equipment deployment, and overall project scheduling. Acted as an industry technical expert to deliver integrated "5G + Aviation" solutions. Ensured efficient and compliant execution of C919 test flights, contributing to the commercialization of China's first domestically developed large passenger aircraft.
- **Hazardous Weather Prediction via 5G ATG Connectivity:** Developed a real-time data integration platform utilizing 5G air-to-ground (ATG) connectivity, incorporating data from aircraft sensors and passenger mobile devices. Trained meteorological prediction models using machine learning to enable precise forecasting of severe weather events such as convective storms and wind shear.

2018.07—2020.07

Spring Airlines
Product Manager

Responsibilities

- Led business consulting and process optimization initiatives by identifying key pain points and promoting the standardization of IT systems to enhance overall operational efficiency.
- Conducted regular industry trend analysis and competitive benchmarking to support the development of the company's mid-to-long-term strategic planning. Promoted data-driven decision-making and the application of emerging IT technologies in civil aviation.

Project Experience

- **EFB (Electronic Flight Bag) Project:** Served as project lead to successfully implement the first paperless flight dispatch system in China's civil aviation industry. The system significantly improved pre-flight safety checks and emergency response for flight changes, saving over RMB 1.5 million annually.
- **Flight Operation Quality Analysis Platform:** Led the design and implementation of an intelligent analysis platform covering airports, routes, and flights. Achieved deep integration of quality data from the airline, airport, and air traffic control. Enhanced flight quality efficiency by over 20%.
- **COMAC Flight Operation Project:** Delivered a customized flight operation management platform for the COMAC flight test center. Covered full-process functions including intelligent scheduling, dispatch, flight monitoring, and quality evaluation. The system empowered data-driven flight quality analysis and significantly improved test flight execution efficiency and quality control, earning high client praise.
- **Spring Airlines WMS Project:** Built a full-process Warehouse Management System for Spring Airlines, covering in-flight consumables from warehouse to aircraft, onboard sales, and return. The system enabled real-time tracking, intelligent handover based on sales models, and self-service inventory checks. It streamlined warehousing, dispatch, and reconciliation, greatly improving supply efficiency and accuracy.

2015.07—2018.07

Shenzhen Airlines
Product Manager

Responsibilities

- Independently led product development from concept to launch and drove product promotion initiatives.
- Managed R&D teams, conducted business analysis, and provided continuous support across both business and technical domains.

Project Experience

- **CDM Airport Operations Platform:** Responsible for integrating and managing airport operations data to support subsequent flight coordination and decision-making.
- **FOC Flight Dispatch Platform:** Developed and optimized features such as dispatch process monitoring, flight status tracking, and anomaly alerts, providing tailored solutions for clients.

Research projects

Multimodal Visual Perception and Semantic Decision-Making 2025.3 - Now

Designed an intelligent driving video analysis system to address the challenge of real-time environmental perception and semantic interpretation in autonomous driving. Leveraged YOLOv8 for object detection, SORT for multi-object tracking, and on-device LLMs for generating contextual driving alerts. Enabled real-time decision-making in complex traffic scenarios through multimodal sensor fusion and semantic understanding.

Sanding Robot Optimization Using DDPG Algorithm 2024.9 - 2024.12

https://github.com/Xiong-Junlin/Reinforcement-Learning-Project---Sanding-Task

Aimed to enhance robotic surface treatment precision in unstructured environments by applying deep reinforcement learning. Developed a simulation-based control model using the DDPG algorithm with PyTorch to optimize a robotic arm's sanding behavior. The model achieved stable learning performance and effective task completion in variable surface geometries.

Sensor Fusion of GPS and IMU Using Extended Kalman Filter 2024.9 - 2024.11

https://github.com/Xiong-Junlin/Sensor-Fusion

Addressed the low accuracy and instability of single-sensor localization in autonomous vehicles by implementing a sensor fusion model. Used Python (NumPy) and the KITTI dataset to build an Extended Kalman Filter that effectively integrated GPS and IMU data. The model significantly improved localization precision and path estimation in simulated urban driving.

Path Planning Algorithms for Mobile Robots Using RRT 2024.5 - 2024.8

https://github.com/Xiong-Junlin/Path-Planning-Using-RRT-Algorithm

Targeted the problem of obstacle-rich navigation in mobile robotics. Designed and implemented a path planning module using the Rapidly-Exploring Random Tree (RRT) algorithm in Python, deployed in the Webots simulator. Enabled robots to autonomously compute collision-free, optimized paths in dynamic environments.

Modeling, Simulation, and Control of Quadcopters 2024.9 - 2024.10

https://github.com/Xiong-Junlin/Modelling-Simulation-and-Control-of-a-Quadcopter

Focused on understanding flight dynamics and control of quadcopters for aerial robotics applications. Analyzed academic literature and used MATLAB to simulate quadcopter behavior in a controlled lab setting. Developed system models, control algorithms, and stability evaluations for multi-rotor flight control.